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Random Forest Modeling for Survival Analysis of Cancer Recurrences

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***Abstract*— The recurrence of breast cancer is a prevailing problem that decreases the quality of patients' lives, creates high burdens on the healthcare system, and impacts the well-being of society. Advanced sensing provides an unprecedented opportunity to increase information visibility and characterize patterns of event occurrences. However, few, if any, of previous works have investigated survival analysis of breast cancer recurrences based on large amount of data readily available in the health system. There is a dire need to leverage data to decipher important factors that play a role in the recurrence of breast cancer. This paper presents an ensemble method of random survival forest for time-to-event analysis of breast cancer recurrences in the surveillance, epidemiology, and end results (SEER) data from year 1973 to 2015. Our model characterizes the survival function among patients with and without recurrences of breast cancer. Ensemble models are constructed via sampling and bootstrapping into the big data. Experimental results show that the age when cancer recurrence happens and time-between-recurrences approximately follow the Gaussian and exponential distributions with the means of 61.35 $\pm$ 14.03 and 2.61 years, respectively. In addition, the results show age, surgery status, stage of tumors, and histological grade are significant factors that influence the probability of breast cancer recurrences. The proposed survival analysis approach shows strong potentials to help healthcare practitioners in prognosis, treatment, and decision-making of breast cancer recurrences.**

Keywords: survival analysis, time-to-event data, random forest, breast cancer, cancer recurrences, ensemble modeling.

I. INTRODUCTION

Breast cancer is a common type of cancer and is the second leading cause of death among women in the US. It is estimated that approximately 268,600 women are diagnosed with breast cancer and 42,260 deaths occur in 2019 [1]. A large number of surviving patients still face the probability of breast cancer recurrence. This leads to the increasing healthcare cost and decreases the patients' quality of life during the post-surgery period [2], [3]. Recurrence is the come back of breast cancer after the treatment, which creates big burdens on the healthcare system. The recurrence of breast cancer is a complex problem and may be due to numerous factors such as age, tumor status and histology results [4], [5]. Currently, healthcare data about the breast cancer recurrences are gathered from different sources (e.g., single-institution patient cohorts and clinical trials), providing useful information about the breast cancer recurrences [6], [7]. However, these raw data are not fully utilized to identify the vital factors in breast cancer recurrences due to the dearth of public information about risk patterns [8], [9], [10]. Therefore, there is an urgent need to develop analytical methods that leverage the big data to identify key risk factors that impact the recurrences of breast cancer.

The methods of survival analysis have been widely utilized for the analysis of time-to-event data. Cox proportional hazards (PH) model is one of the most common regression modeling frameworks for survival analysis [11], [12]. However, this hazard model relies on the restrictive assumption that the ratio between hazards is constant over time. Because of the long-term treatment period for breast cancer patients, the PH assumption is often violated, leading to a poor prediction model [13]. Furthermore, other survival models are often parametric, which limits their ability to model nonlinear effects and complex interactions among variables. On the other hand, constructing ensembles of predictive models increases the prediction performance. The ensemble method of random survival forest (RSF) injects randomization to the survival model. The random forest bootstraps samples into the base learning process and selects a subset of variables as a candidate for splitting each node of the tree [14]. Note that random sampling and model splitting at each node in conjunction with averaging over trees leads to a significant reduction of prediction error [15], [16].

Healthcare surveillance involves a continuous and systematic collection of health-related data, which presents a new area of research especially for the care of breast cancer survivors. Several organizations (e.g., American Cancer Society) have provided guidelines for clinical surveillance or follow-up care of patients after they receive breast cancer treatments [17]. By examining the variation of the hazard function over time, analytical models determine when the event of breast cancer is more likely or less likely to occur, thereby providing a general profile of risk over time. However, registries just capture information about the healthcare outcomes (e.g., diagnostic results and deaths) due to the disease or other causes. Few, if any, studies used the extensive Surveillance, Epidemiology, and End Results (SEER) data to model the variations in the hazard function and time-between-recurrences for breast cancer patients in the US [18], [19]. There is an urgent need to leverage the big data from different sources to help practitioners in prognosis, treatment, and decision-making of breast cancer recurrences.

The proposed research is aimed at filling these gaps by developing the random forest model for survival analysis of breast cancer recurrences. Note that RSF, as a nonparametric method, is capable of delineating nonlinear effects and interactions. The proposed approach includes a suite of analytical tools such as data prepossessing, variable selection, risk modeling, and data visualization. A baseline model is

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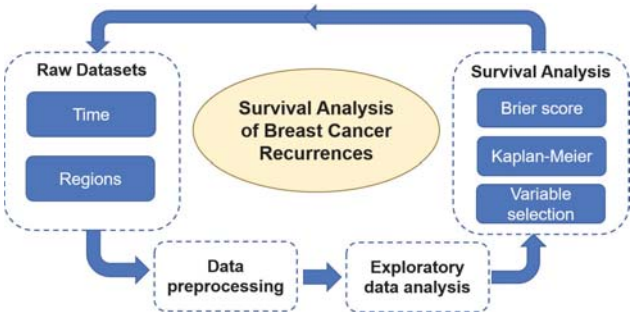

Fig. 1. Flow diagram of the research methodology

established by estimating the survival function for a patient with and without recurrence of breast cancers using features such as age, marital status, sequence number, histological grade, tumor size, and treatment status.

The remainder of this paper is organized as follows: Section II presents the research methodology for survival analysis of time-to-event data. Section III describes experimental results of RSF modeling of cancer recurrences in the SEER database, and Section IV concludes the present paper.

II. RESEARCH METHODOLOGY

As shown in Fig. 1, the proposed methodology consists of the following steps: 1) data integration, 2) data preprocessing, 3) exploratory data analysis and 4) survival analysis of breast cancer recurrences.

A. Data Preprocessing

The SEER program, supported by the U.S. National Cancer Institute (NCI), collects data from tumor registries and covers 14% to 25% of the U.S. population. The breast cancer data include more than one and half million records from different regions of the U.S. from year 1973 to 2015. These registries' data are vital to analyzing and reporting the evolving burden of breast cancer in the population.

There are a total of 1,631,572 entries in SEER database. Each entry has the same size with coded cancer-related variables such as year of diagnosis, month of diagnosis, grade, stage, sequence, etc. Some important variables related to this work are listed in Table I. Note that the patient ID is utilized to uniquely identify a patient in the SEER database. Based on the patient ID, we create a tensor structure (i.e., 3-dimensional data structure of patient, variables, and breast cancer recurrences) for all patients with re-retry records. Note that in each layer, we store data of all records of the same patient. For example, if the patient has breast cancer recurrences (i.e., they will have multiple records), there will be more than one record stored in the corresponding layer.

B. Survival Analysis

In many breast cancer studies, the key outcome under investigation is the time-to-event. Survival analysis is a powerful tool for characterizing the patterns of event occurrence to reveal key properties of breast cancer.

TABLE I
THE LIST OF SOME IMPORTANT VARIABLES IN THE SEER DATABASE

Variable	Description
Patient ID	Used in conjunction with SEER registry to uniquely identify a patient
Age	Represents the age of the patient at diagnosis
Year	Describes the year of diagnosis
Month	Documents the month of diagnosis
Sex	Identifies the sex of the patient at diagnosis
Race	Shows the race information of the subject
Marital status	Identifies patient's marital status
Grade	Indicates what the cells look like and how fast the cancer grows
Stage	Identifies the stage of the tumor
Surgery	Documents if the surgery is performed
Primary site	Identifies the site in which the primary tumor originated
Histology	Defines the cell type of the tumour
Sequence number	Describes the number and sequence of all reportable malignant, in-situ, benign, and borderline primary tumors, which occur over the lifetime of a patient
Laterality	Describes the side of a paired organ or side of the body on which the reportable tumor originated
Behavior	Identifies the behavior of the tumor
Diagnostic	Records the best method used to confirm the presence of the breast cancer

Let $T \geq 0$ denotes survival time as a random variable. The survival function is the probability density function of a patient survives more than time t and is written as:

$$S(t) = P(T > t) \tag{1}$$

The hazard function is the instantaneous rate of the events occurrence for patients who are surviving at time t. The hazard function and cumulative hazard function (CHF) are defined respectively as:

$$h(t) = \lim_{\Delta t \to} \frac{P(t \leq T < \Delta t | T \geq t)}{\Delta t} \tag{2}$$

$$H(t) = \int_0^t h(x)dx \tag{3}$$

The cumulative function in Eq. (3) is calculated based on survival function as:

$$S(t) = e^{-H(t)} \tag{4}$$

Survival and hazard functions can be estimated using a nonparametric method such as the Kaplan-Meier estimator or parametric methods such as exponential, Gamma or Weibull functions. Time-to-event analysis is aimed at the estimation of survival function and then the comparison of survival times between different groups.

An alternative to building a single survival tree is to build many survival trees, where each tree is constructed using a random sample of the data and many trees are further combined to estimate the survival function. This leads to a more accurate prediction in comparison with traditional survival methods such as PH or Cox PH models. The detailed steps of the RSF method are as follows:

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- Draw *B* bootstrap samples from the data. Note that each bootstrap sample leaves out a portion of the data, called out-of-bag (OOB) data.
- Develop a survival tree for each bootstrap sample. At each node of the tree, we will randomly select *p* candidate variables and then use the log-rank test to maximize the survival difference between child nodes.
- Terminal node for each tree should have no less than $d_0 > 0$ unique deaths.
- Calculate a CHF for each tree and then take the average to obtain the ensemble CHF. The OOB data are used to calculate the prediction error of ensemble models.

RSF works with right-censored data, which occur when a patient does not experience the event of interest for the duration of the study. The key components of RSF method include the spread of survival trees as well as the construction of ensemble CHF. Note that a bootstrap dataset of size n is built by drawing n samples with replacement from the original dataset. Here, nodes are divided into two groups as the survival trees grow recursively based on the log-rank test. An optimal split for a node is obtained by searching all variables and maximizing the differences of survival predictions between child nodes. As the tree grows, each node in the tree becomes homogeneous and is populated by cases with similar survival functions.

The convergence is achieved when there is no new child can be generated based on the criterion that each node should include at least $d_0 > 0$ deaths. Let $(x_{i,h}, \gamma_{i,h})$ be the survival time for the patient *i* in the node *h*. A patient is right-censored at the time $x_{i,h}$ if $\gamma_{i,h} = 0$, otherwise the patient has the recurrence of the breast cancer at $x_{i,h}$. Let $x_{1,h}, ..., x_{I,h}$ be the recurrence times at node h, $r_{i,h}$ and $R_{i,h}$ be the number of recurrences and the number of risky patients at time $x_{i,h}$. The CHF estimate of *h* can be achieved via Nelson-Aalen estimator as follows:

$$\hat{H}_h(x) = \sum_{x_{i,h} \leq x} \frac{r_{i,h}}{R_{i,h}} \tag{5}$$

For the patient *i*, we have a d-dimensional covariate $\mathbf{c}_i$. Let $H(x|\mathbf{c}_i)$ denote the CHF for patient *i*. Due to the binary nature of survival tree, $\mathbf{c}_i$ locates in terminal node *h* and CHF based on Nelson-Aalen estimator for $\mathbf{c}_i$ will be:

$$H(x|\mathbf{c}_i) = \hat{H}_h(x) \quad \text{if } \mathbf{c}_i \in h \tag{6}$$

The CHF in Eq.(6) is calculated based on a tree. RSF as an ensemble method needs to average the results of *B* trees. Here, each tree is grown by bootstrapping samples independently. Let's denote $\delta_{i,b}$ as an indicator variable which has a value 1 if patient *i* is in OOB set and zero if patient *i* is not included in OOB set. Let $H_b(x|\mathbf{c}_i)$ denote the CHF for the bth tree of the patient *i*. Therefore, the CHF for patient *i* based on OOB ensemble and for the bth bootstrap sample is:

$$H_e^*(x|\mathbf{c}_i) = \frac{\sum_{b=1}^{B} \delta_{i,b} H_b(x|\mathbf{c}_i)}{\sum_{b=1}^{B} \delta_{i,b}} \tag{7}$$

The $H_e^*(x|\mathbf{c}_i)$ is an average of bootstrap samples for a patient *i* in the OOB dataset. Also, we can reformulate the above equation by dropping OOB data as:

$$H_e(x|\mathbf{c}_i) = \frac{1}{B} \sum_{b=1}^{B} H_b(x|\mathbf{c}_i) \tag{8}$$

Note that the total number of recurrences equals to the sum of the CHF over $(x_i, \mathbf{c}_i)$. We can define the average CHF over the time x_i conditioned on the $\mathbf{c}_i$ as the mortality that provides an estimate of the number of recurrences under the null hypothesis of homogeneous survival behavior. The ensemble mortality for a patient *i* can be written as:

$$D_i = E_i(\sum_{j=1}^{n} H(x_j|\mathbf{c}_i)) \tag{9}$$

Based on ensemble characteristics of the survival method, we can define the ensemble mortality for patient *i*:

$$\hat{D}_{e,i} = \sum_{j=1}^{n} H_e(x_j|\mathbf{c}_i) \tag{10}$$

Also, the OOB ensemble mortality can be expressed as $\hat{D}_{e,i}^* = \sum_{j=1}^{n} H_e^*(x_j|\mathbf{c}_i)$. The concordance statistic is commonly used to measure the error in survival analysis which is similar to the goodness-of-fit for binary outcomes or the area under a receiver operating characteristic (ROC) curve. C-index (C_e) utilizes the predicted outcome calculated based on OOB ensemble CHF. Let $x_1^o, ..., x_m^o$ be the predefined time points. If we have two cases *i* and *j*, then *i* has a worse predicted outcome than *j*:

$$\sum_{k=1}^{m} H_e^*(x_k^o|\mathbf{c}_i) > \sum_{k=1}^{m} H_e^*(x_k^o|\mathbf{c}_j) \tag{11}$$

We leverage Eq. 11 to calculate C-index for OOB data and consequently define the prediction error as $1 - C_e$. Note that this prediction error has a value between zero and one. The Brier score is a popular metric of prediction error in survival analysis, which can be written as:

$$B(x) = \frac{1}{N} \sum_{k=1}^{N} \{ \frac{(0 - \hat{S}(x|\mathbf{c}_i))^2}{\hat{S}'(x_i)} (\delta_{x_i \leq x, \gamma_i = 1}) + \frac{(1 - \hat{S}(x|\mathbf{c}_i))^2}{\hat{S}'(x_i)} (\delta_{x_i > x}) \} \tag{12}$$

where $\hat{S}'$ is the estimated survival function from the Kaplan Meier method.

III. EXPERIMENTAL RESULTS

We leveraged the SEER dataset from various regions and over different time periods to study and characterize recurrent events among the patients who are initially diagnosed with breast cancer in the U.S. from year 1973 to 2015.

As shown in Table II, the combined dataset includes 1,631,572 entries where 1,620,724 are female. In the SEER

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TABLE II

A SUMMARY OF SEER DATASETS FOR BREAST CANCER

Category	Group	Count
Gender	Male	10848
	Female	1620724
Age	Below 40	102464
	41-50	302175
	51-59	388714
	61-70	387000
	71-80	294386
	Over 80	156734
	Unknown	99
Marriage Status	Single (never married)	200438
	Married (including common law)	895249
	Others	461500
	Unknown	74385
Race	White	1342756
	African American	158692
	American Indian	7615
	Asian	76463
	Others	39553
	Unknown	6493
Surgery	had surgery	1506046
	did not have surgery	113112
	Unknown	12414

TABLE III

THE NUMBER OF RECURRENCES AMONG PATIENTS DIAGNOSED WITH BREAST CANCER

Number of cancer recurrences	Patient count
2	116764
3	5679
4	344
5	20
6	2

database, there are 252,053 re-entries records. Note that one patient can have multiple breast cancer recurrences, the re-entries are from 122,809 patients in 18 different regions. Among these patients, there are 102,282 white, 11,285 African American, 541 American Indian, 7,360 Asian, and 1,341 other races. Results show that the recurrence rate of breast cancer is about 7%.

As shown in Table III, the number of recurrences varies from 2 to 6. Note that the patient count hits the peak of 116,764 when the number of cancer recurrences is 2, while the patient count hits the minimum of 2 occurs when the number of cancer recurrences is 6. We performed the recurrence analysis based on ethnicity. Results show that most recurrences happen among White patients. Quantitatively, 83% of breast cancer recurrences occur among White patients. African American and Asian are the other ethnicities with a high rate of cancer recurrences. The rate of recurrences are 9% and 6%, respectively. The analysis is also performed based on the fact that whether a patient has a Hispanic origin. The results show that Hispanic patients have the 7% chance of recurrences, which is the same as the percentage of recurrences in the dataset.

Furthermore, we extract the age of patients when the event of breast cancer occurred and reoccurred. To perform the survival analysis for recurrence data, there is a need

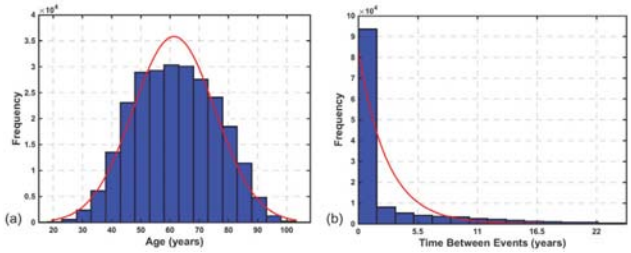

Fig. 2. (a) The age distribution of cancer recurrences; (b) The distribution of time-between-events for breast cancer patients

to extract the time between events for the patients with recurrences. As shown in Fig. 2 (a) and (b), the average age for cancer recurrence is 61.35 with a standard deviation of 14.03 years. Also, we look into the age distribution when the breast cancer reoccurs. From the age 20 to 50, the rate of breast cancer recurrence increases. However, when the patient is between the age of 51 and 70, the rate of recurrence tends to be constant. After the age of 70, the rate of cancer recurrence decreases. In addition, the time between recurrences approximately follows an exponential distribution with mean 2.61 years. Note that the majority of recurrences occur within the first 3 years after a patient is diagnosed with breast cancer.

The existence of a large number of variables in the SEER dataset makes the computational cost of survival analysis expensive. We utilize the random forest to build the survival model with variables in the SEER dataset. The random forest does not require the explicit specification of the functional form of covariates to the response. Consequently, there is no significance test (e.g., p-value) for variables selection. Random forest determines variables that have a significant impact on survival analysis via the split rule optimization.

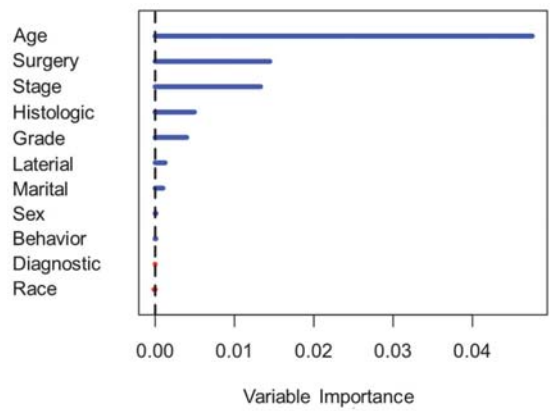

Fig. 3. Random forest modeling and analysis of variable importance to the recurrence of breast cancer

In our experiment, the forest consists of 1000 trees. Model parameters include the size of the terminal node, the number of splits for each variable, and the number of candidate variables at splits. The grid search method is utilized to optimize the size of terminal node, number of splits, and

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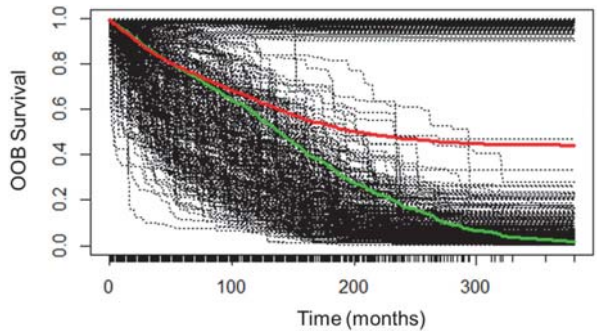

Fig. 4. Kaplan-Meier plots for the patients with breast cancer recurrences (i.e., green line shows the average survival rate for the recurrence group) and for those without breast cancer recurrences (i.e., red line shows the average survival rate for without recurrence group)

candidate variables at splits. The OOB error rate is used as performance metric to choose the optimal survival model.

As shown in Fig. 3, age, surgery history, stage, and histological grade are four important variables that impact the recurrence of breast cancer. We randomly sample data from patients with breast cancer recurrences and from patients did not experience recurrences. It is worth mentioning that if an variable has a bigger importance score, then it has a significant impact on the model output. Those variables with small or negative importance scores tend to be insignificant and are not sensitive to impact the performance of survival models. In the latter, we assume these variables are no significant than random variations to the model. As such, we only count on significant variables with big importance scores to improve the performance of random forest modeling for survival analysis. Fig. 4 shows the OOB survival analysis over a time period of 400 months.

As shown in Fig. 4, the Kaplan-Meier method is utilized to estimate the survival function for each patient. Note that each black line represents a single patient in the dataset, which includes patients without recurrence and those who have experienced the recurrence of breast cancer. Here, the average OOB survival probability for patients without breast cancer recurrence (i.e., the red line) is significantly higher than those who have recurrences (i.e., the green line). Results show that the gap between survival functions is drastically bigger as the time to event increases.

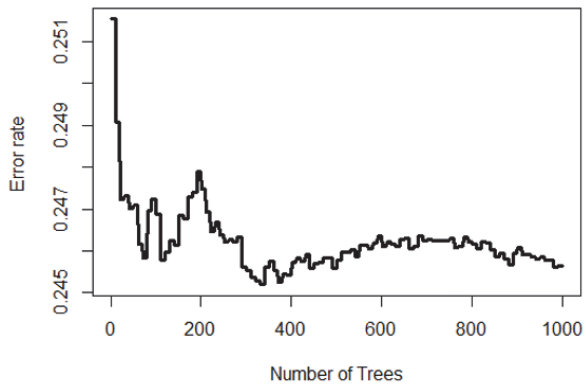

Fig. 5. The variation of OOB prediction errors with respect to the number of trees in the random forest

Fig. 5 shows that a large number of trees is not required for survival analysis. However, to make sure that each variable has a chance to be included in the forest prediction process, we create a large random forest of 1000 trees.

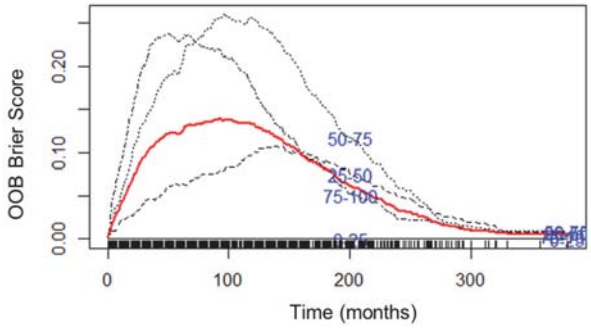

Fig. 6. The Brier score for prediction errors of breast cancer recurrences

Fig. 6 presents the Brier score for prediction errors of breast cancer recurrences with the RSF survival analysis. Brier score compares the survival analysis results of patients with their actual status. Note that a smaller Brier score indicates a more accurate performance of RSF predictive models. Averaging over many trees enables RSF to approximate complex survival functions while maintaining a low prediction error. It is worth mentioning that OOB estimates of prediction error have been proved to be similar to n-fold cross validation estimates. Hence, random forest models integrate the training with validation together to improve the prediction performance.

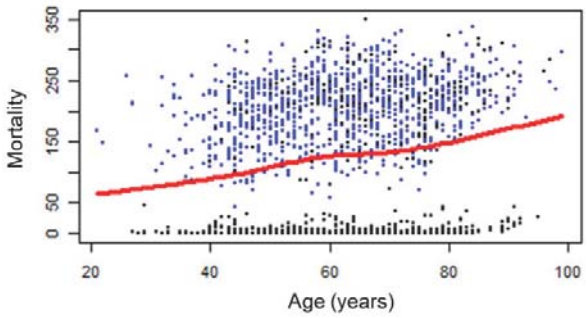

Fig. 7. The number of mortality vs. the age for patients with breast cancer recurrences (i.e., blue dots) and patients without recurrences (i.e., black dots) for the sample size of 2000

Fig. 7 illustrates the impact of the variable of age on mortality for patients with and without the recurrence of breast cancer. Mortality is defined as the expected value for the CHF summed over a specific time and conditioned on a specific covariate. Here, the black dot represents the patients that are diagnosed with breast cancer only for one time, and blue dots show the mortality for the patients who have breast cancer recurrences. Note that the average mortality (i.e., red line) increases with respect to the age.

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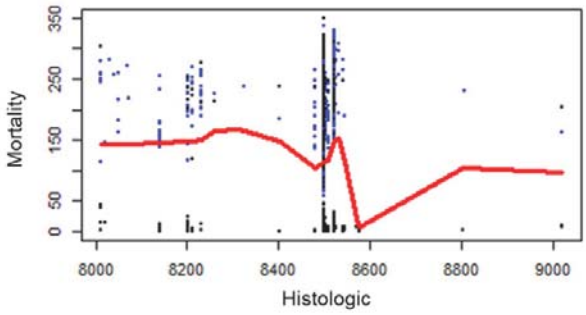

Fig. 8. The number of mortality vs. different histological grades (8000, 8200, 8400, 8600, 8800 and 9000) for patients with breast cancer recurrences (i.e., blue dots) and patients without recurrences (i.e., black dots) for the sample size of 2000

We also perform an analysis of the mortality based on histological grade (i.e., the fourth important factor in breast cancer recurrence). The histological grade determines the microscopic composition of tissue for a patient. Fig. 8 shows that mortality for paragangliomas and glomus tumors are less than other types. Also, the mortality for patients with breast cancer recurrence is greater than those who did not exprience recurrence for different histological grades.

IV. CONCLUSIONS

The recurrence of breast cancer is one of the main concerns for many patients after receiving treatments. Cancer recurrences increase the healthcare cost, create high burdens on the health system, and decrease the patients' quality of life. Advanced sensing and information technology bring big data (e.g., millions of records) about cancer recurrences and provide a unique opportunity to extract useful information for the analysis of cancer recurrences. This paper presents an ensemble method of random survival forest (RSF) to study the impact of a variety of factors such as age, marital status, histological grade on the recurrence of breast cancer. The RSF method helps estimate and predict the survival function by sampling and bootstrapping into the big data with an ensemble model of survival trees. We study the SEER breast cancer dataset, which includes more than one and half million entries to characterize breast cancer recurrences and provide a comprehensive study of the problem. Results show that the recurrence rate of breast cancer is 7% and the number of recurrences varies between 2 and 6. In addition, age, surgery status, stage of tumors, and histological grade are among the most significant factors that impact the recurrence of breast cancer.

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